

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

Fourth Semester of B. Pharm. Examination

University Theory Examination May 2017

PH221 PHARMACOLOGY II

Date: 05.05.2017, Friday

Time: 01:30 p.m. to 04:30 p.m.

Maximum Marks: 80

Instructions:

1. There are three sections in this question paper.
2. SECTION – I comprises of Question 1. Total marks for Section 1 are 20. There are 20 sub-questions (MCQ type). Answers to SECTION – I are to be given in Answer Sheet for MCQ type questions provided to you. Maximum time allotted for SECTION – I is 30 minutes. Answers to SECTION – I must be written during the first 30 minutes of the examination.
3. Answers to SECTION – II and SECTION – III are to be provided in separate Main Answer Books provided to you.
4. Figures to right indicate marks.
5. Draw neat labeled sketches wherever necessary.

15BPH053.

SECTION - I

Q 1 Attempt all questions. Each question is of one mark. 20

1. _____ is an increase in the size of cells resulting in increase in the size of the organ.
[A] Metaplasia
[B] Hyperplasia
[C] Hypertrophy
[D] Atrophy
2. The migration of inflammatory cells out of vessels is the:
[A] Cellular reaction
[B] Exudative reaction
[C] Vascular reaction
[D] Chemotaxis reaction
3. All of the following inflammatory mediators are released from the mast cells except:
[A] Histamine
[B] 5-Hydroxy tryptamine
[C] Prostaglandins
[D] Leukotrienes
4. Which of the following is an example of second generation antihistamines?
[A] Promethazine
[B] Hydroxyzine
[C] Diphenhydramine
[D] Loratadine
5. Which of the following disease condition is not treated by H1 receptor antagonist?
[A] Allergic diseases
[B] Common cold
[C] Preanesthetic medication
[D] Gastroesophageal reflux disease
6. Which of the following drug is 5-HT₃ antagonist and used as anti-emetic?
[A] Ondansetron
[B] Ketanserin
[C] Metoclopramide

- [D] Buspirone
7. 5-Hydroxy tryptamine is involved as a mediator in the following pathological condition:
[A] Delayed hypersensitivity reaction
[B] Variant angina
[C] migraine
[D] Anaphylaxis
8. Dysmenorrhoea is often associated with excess production of the following autacoid by the endometrium:
[A] Platelet activating factor
[B] 5-Hydroxytryptamine
[C] Bradykinin
[D] Prostaglandin
9. Montelukast blocks the action of the following autacoid:
[A] Leukotriene B₄
[B] Leukotriene C₄/D₄
[C] Platelet activating factor
[D] Prostacyclin
10. Codeine is used clinically as:
[A] Analgesic
[B] Antitussive
[C] Antidiarrhoeal
[D] All of the above
11. The following class(es) of drugs are clinically beneficial in bronchial asthma:
[A] Leukotriene (cys LT₁) receptor antagonists
[B] Histamine H₁ receptor antagonists
[C] Platelet activating factor (PAF) antagonists
[D] All of the above
12. The first choice drug for nonsteroidal anti-inflammatory drug associated gastric ulcer is:
[A] Omeprazole
[B] Misoprostol
[C] Sucralfate
[D] Ranitidine
13. Which of the following antacid is having highest acid neutralising capacity (ANC)?
[A] Magnesium Trisilicate
[B] Magnesium Hydroxide
[C] Aluminium Hydroxide
[D] Calcium carbonate
14. Which of the following anti-emetic is belong to the category of Substance P antagonist?
[A] Metoclopramide
[B] Ondansetron
[C] Aprepitant
[D] Domperidone
15. The following laxative lowers blood ammonia level in hepatic encephalopathy:
[A] Bisacodyl
[B] Liquid paraffin
[C] Lactulose
[D] Magnesium sulphate

16. Which of the following is used in treatment of acute iron poisoning administered by oral route?
 [A] D-penicillamine
 [B] Deferiprone
 [C] Desferrioxamine
 [D] Disodium EDTA
17. Which of the following chelating agent is contraindicated in the treatment of cadmium poisoning?
 [A] Dimercaprol
 [B] D-Penicillamine
 [C] Disodium EDTA
 [D] Calcium Disodium edetate
18. What is gene therapy ?
 [A] Pre-clinical testing for inherited diseases in newborns
 [B] Treatment of diseases caused by genetic defect
 [C] Genetic engineering using recombinant DNA technology
 [D] Cancer treatment using in vitro cultured stem cells
19. The first gene therapy used cells altered outside the recipient's body and is called ____ gene therapy.
 [A] In-situ
 [B] Ex-vivo
 [C] In-vivo
 [D] vaccine
20. The following eicosanoid is generated through the lipoyxygenase pathway:
 [A] Prostaglandin E2
 [B] Thromboxane A2
 [C] Leukotriene C4
 [D] Prostacyclin

SECTION – II

- Q 2** Attempt any TWO of the following
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|----------|--|-----------|
| A | Differentiate: Cimetidine and Ranitidine | 05 |
| B | Justify the use of prostaglandins in cardiovascular and gynecological conditions | 05 |
| C | Comment: Second generation antihistamines are preferred over first generation antihistaminics. | 05 |
- Q 3** Attempt any TWO of the following
- | | | |
|----------|---|-----------|
| A | Write a note on chelating agent. | 05 |
| B | What are different considerations for selection of a delivery system?
Discuss different non viral vectors used for gene therapy. | 05 |
| C | Write a note on Apoptosis. | 05 |

SECTION – III

- Q 4** Attempt any **FOUR** of the following
- A** Mention H₁ and H₂ receptor mediated action of histamine. What is triple response caused by histamine? **05**
- B** Classify anti-emetics. Explain mechanism of action of each class. **05**
- C** What is inflammation? Describe the role of mediators of inflammation. **05**
- D** Enumerate drugs used in treatment of bronchial asthma and give mechanism of action of beta agonist in asthma. **05**
- E** Enlist leukotriene receptors. Describe pharmacology of leukotrienes. **05**
- F** Enlist the causes of cell injury. Explain biochemical mechanism of cell injury due to ATP depletion. **05**
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- Q 5** Attempt any **FOUR** of the following
- A** Discuss different types of cellular adaptations with suitable examples. **05**
- B** Enlist autocooids. Write pharmacological actions of 5-HT. **05**
- C** Classify anti-ulcer drugs. Explain mechanism of action of H₂ receptor antagonist and proton pump inhibitors **05**
- D** Write a note on antitussives and expectorants. **05**
- E** Define and classify laxatives & purgatives. Write their therapeutic uses. **05**
- F** Write a note on prokinetic agents. **05**
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